

- 1 (a) Woodlice are small invertebrates that live in cool, damp places under wood and stones.
- Some students determined the respiration rate of woodlice at different temperatures.

Fig. 1.1 shows a single woodlouse.

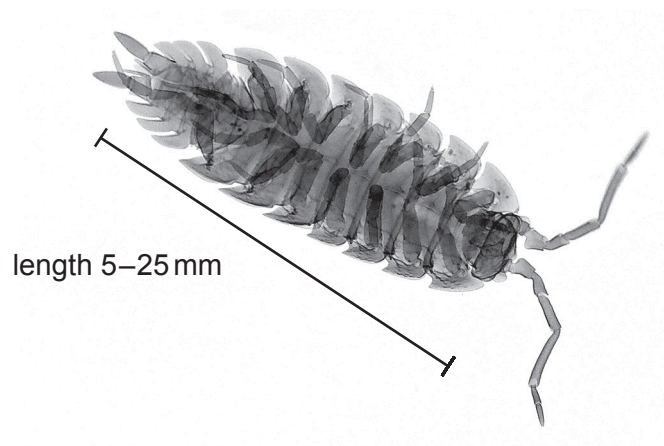


Fig. 1.1

The students tested the hypothesis:

For each 10°C increase in temperature the volume of oxygen absorbed will double.

Fig. 1.2 shows the respirometer that the students used.

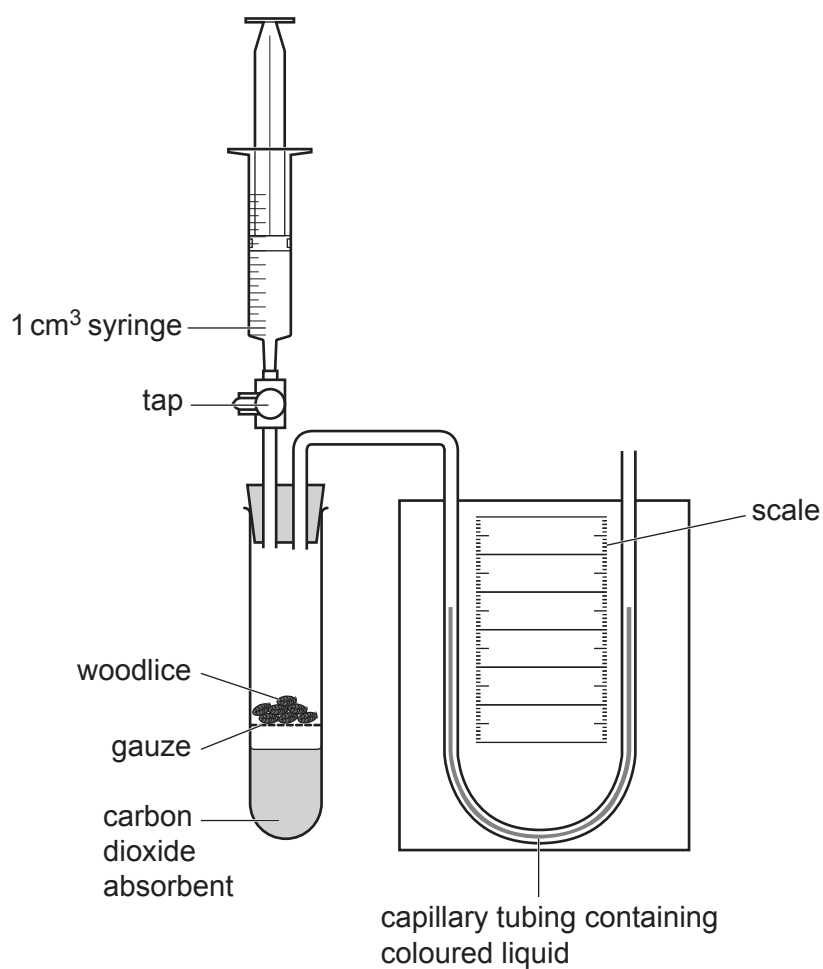


Fig. 1.2

- (i) State the independent variable and the dependent variable in this investigation.

independent variable

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dependent variable

.....

[2]

- (ii) The students decided to measure oxygen uptake for 5 minutes using a range of temperatures from 5 °C to 35 °C.

Suggest why the students chose this range of temperatures **and** suggest a suitable interval that the students should use.

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..... [2]

- (iii) State a suitable control for this investigation.

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..... [1]

- (iv) State why the carbon dioxide absorbent solution was placed in the tube.

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..... [1]

- (v) Suggest **one** variable in this investigation that **cannot** be standardised.

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..... [1]

- (b) (i)** Describe a method by which the students could set up the respirometer **and** use it to test their hypothesis:

For each 10°C increase in temperature the volume of oxygen absorbed will double.

Your method should be set out in a logical order and be detailed enough to let another person follow it.

[7]

- (ii) Each 1 cm length of the capillary tubing containing coloured liquid in Fig. 1.2 has a volume of 10 mm^3 .

Describe how the students could use their results to calculate the rate of respiration for each temperature from their results.

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..... [3]

- (iii) Use the axes in Fig. 1.3 to show the expected shape of the curve if the hypothesis is correct.

For each 10°C increase in temperature the volume of oxygen absorbed will double.

Include the labels and units for each axis.



Fig. 1.3

[3]

- (c) The students used the same apparatus to determine the output of carbon dioxide by the woodlice at 35°C . The students used this result and the result for oxygen uptake at 35°C to calculate the respiratory quotient (RQ).

The RQ was calculated as 0.95.

State what conclusion can be made from this RQ value.

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..... [1]

[Total: 21]